GREAT IDEA #6 OF 7

Memory Hierarchy

Programmers want memory to be:

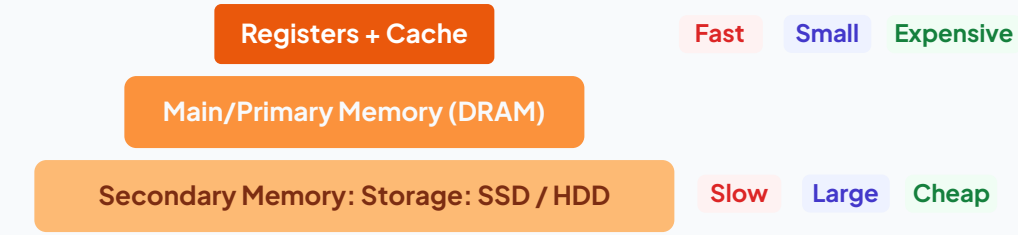
Fast
Affects Performance

CPU can stall if memory not fast enough.

Large in Capacity
Limits the size of the problem

Cheap
Memory Cost is a major factor

The Layered Pyramid



Interpretation: The closer to the top, the faster and more expensive per bit. The wider the base of the layer, the larger the capacity.

↕ Inverse trade-off between speed/cost and capacity

Core Idea: No single technology is simultaneously fastest, largest, and cheapest.

The Cache creates an illusion for the programmer: It keeps frequently accessed data in expensive, fast memory and less-used data in cheaper storage, making main memory appear almost as fast as cache and as vast/cheap as disk.

Details: in Chapter 5